Check for updates

Received: 14 July 2022 | Revised: 18 March 2023 | Accepted: 23 June 2023

DOI: 10.1111/exsy.13394

ORIGINAL ARTICLE

Expert Systems WILEY

Concept drift detection and adaption framework using optimized deep learning and adaptive sliding window approach

Ketan Sanjay Desale | Swati V. Shinde

Department of Computer Engineering, PCET's Pimpri Chinchwad College of Engineering, Pune, India

Correspondence
Ketan Sanjay Desale, Department of Computer Engineering, PCET's Pimpri Chinchwad College of Engineering, Sector-26, Pradhikaran, Nigdi, Pune, India.
Email: desaleketan.10@gmail.com

Abstract

Concept drift in online streaming data is a common issue due to dynamic smart systems, which results in system failure or performance degradation. Though there are several traditional approaches for handling the streaming data, they failed to handle the concept drift imposing the need for developing an adaptable approach to managing the dynamic IoT streaming data. Therefore, in this research, a new method is proposed for handling the concept drift issues in online data streaming. This research develops the dynamic streaming data analytic framework based on the optimized Deep CNN and optimized adaptive and sliding window (OASW) approach that effectively addresses both memory and time constraints. An optimized Deep CNN classifier is employed as a base classifier for offline learning, which is developed through hybridizing the proposed Desale's aggressive hunt optimization (AHO) algorithm with a Deep CNN classifier for tuning the optimal parameters of the classifier. An optimized adaptive and sliding window is utilized in this research to adapt the pattern changes in the data streams, which effectively handles the concept drift. The experimental analysis reveals that the proposed methods outperform the conventional methods considered for the analysis in terms of specificity, sensitivity, accuracy, *F*1 score, and the precision score of 96.65%, 97.77%, 98.63%, 98.1487%, and 98.4469%, respectively.

KEYWORDS
concept drift, deep CNN, machine learning, optimization

1 | INTRODUCTION

Concept drift in machine learning describes how an underlying problem's input–output linkages evolve. Any type of alteration to the data could occur during concept drift. When the change has some degree of temporal consistency, meaning that data gathered over a while consistently indicate the same relationship and that this relationship changes gradually over time. IoT plays an inevitable task in today's world, and the concept drift presence in the IoT device reduces the anomaly detection models performance which causes security problems. An illustrative example of concept drift is spam detection, where the system that receives this spam represents a malicious attack on the computer, which needs to be resolved by the detection of the concept using efficient methods. Concept drift affects the quality of predictions and sustains high predictive performance by using concept drift detection models to detect and manage distributional changes. Ideally, models for idea drift detection are sensitive enough to pick up drift with just a brief delay. To prevent false positives and be trusted, concept drift detection also is resilient against minor input changes (Haug & Kasneci, 2021). The challenges in the research are:

 © 2023 John Wiley & Sons, Ltd.

- Time and memory constraints: Challenges in processing and storing data from IoT devices which are low cost and for meeting real-time processing requirements, the data analytics speed must be higher than data collection/generation time else causes IoT service unavailability or system failure.
- Concept drifts detection: Two main challenges of drift detection are many causing factors and multiple types of drifts in IoT systems
- Concept drift adaptation: It is a difficult task to adapt the new data patterns and to handle the observed drift effectively after drift detection.
- Noise: Challenges in protecting the multiple types of data, such as embeddings, extracted features, and generated instances from the noise.
- Catastrophic forgetting: Protecting important information based on a memory management strategy and transferring knowledge and skills for future tasks are difficult.

Any type of drift could be identified and if the drift occurred does not occur before the deep CNN update the information about drift and further proceeds, which helps in detecting the concept of drift effectively. Hence, this research is capable of adapting new patterns even in the noise existence and proposed model could work well, where the memory is also optimized using the optimization proposed in the research, which tends to resolve the above challenges.

In this research, an innovative drift adaptive framework for online streaming data analytics is proposed. The proposed framework is comprised of the Deep CNN network for offline learning, which is optimized by the proposed Desale's aggressive hunting optimization (AHO). Further, the proposed model utilizes optimized adaptive and sliding windows (OASW) for adaptation of concept drift during the incremental learning stage. The standard dataset, such as CHB-MIT Scalp EEG Database (database 1) and Siena Scalp EEG Database (database 2) are utilized to analyse the proposed drift detection and adaption model.

1.1 | Proposed Desale's aggressive hunting optimization

Desale's aggressive hunting optimization is a nature-inspired algorithm, which is developed by integrating the hunting characteristics of the sea lion (Search agents) and the coyote. The hunting characteristic of the search agents is improved by the communicative behaviour of the coyote. The hybridization of these characteristics provides better efficiency to the classifier.

1.2 | AHO-based deep CNN

The concept drift detection and adaption is accomplished through the Deep-CNN classifier, where the hyper-parameters of the classifiers are tuned optimally by the proposed Desale's aggressive hunting optimization. This tuning helps in detecting the target variable that causes concept drift.

The organization of the manuscript follows: the literature review with the concept drift detection model need is described in Section 2. The proposed model of drift detection and adaption model is elucidated in Section 3 and the result analysis is described in Section 4. Eventually, the conclusion is depicted in Section 5.

2 | BACKGROUND AND LITERATURE REVIEWS

The requirements of advanced concept drift prediction and adaptive method and the major challenges associated with the conventional methods are detailed below.

Darem et al. (2021) presented sequential deep learning and attained enhanced efficiency and accuracy concerning the frequency and updating time. Yet the sequential deep learning model lacked labelled samples. Korycki and Krawczyk (2020) implemented Robust Restricted Boltzmann Machine for Drift Detection high detection rate and robustness against the possible adversarial attack. Liu et al. (2020) used Fuzzy Distance Estimations and provided more reliable results. Computation costs were increased dramatically with high-dimensional data sets. Xu et al. (2020) used the time-series problems to attain the best security level. Moreover, the research was confirmed on static data. Casado et al. (2021) presented Concept-Drift-Aware Federated Averaging (CDA-FedAvg) adapted changes in non-stationary situations. Nagar et al. (2020) used Multivariate Hoeffding's bound approach this approach does not over-segment and precisely located the temporal boundaries. But did not mention the problems where videos were captured at an extremely low temporal solution. Demsar and Bosnic (2018) presented a model explanation for determining concept drifts and effectively handled redundancies, disjunctions, and noise. The main limitations were the high level of parameterization and the required computational time. Guo et al. (2021) utilized adaptive deep neural networks. The fitting capacity of the model to the data distribution is improved but failed to manage high dimensional nonlinear problems. Abbasi et al. (2021) initiated an El Stream that detected concept drift using both machine learning techniques and ensemble techniques. Both artificial data and real data were used in real-life applications and this

14680394, 2023, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/exsy.13394 by Universidad del Bio Bio, Wiley Online Library on [18/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

method helped in big data analysis but the computational complexity is high. Yu et al. (2019) developed an adaptive online learning model that used a drift map concept to adjust the forgetting factor of an extreme learning machine that provided an effective solution in a non-stationary environment for constructing the online learning models but still automated determination of optimal parameters method is not implemented.

Priya and Uthra (2021) implemented the Statistical Test of Equal Proportions—Kernel Extreme Learning Machine model and found an efficient tool for detecting email spam. However, the accuracy value was decreased over the sample of the recent window. Yuan et al. (2021) presented a neural network based on the Adadelta optimizer model as an effective method and classified highly imbalanced streaming data. But this method had not used feature selection and clustering techniques, for reducing computational complexity.

There is a rapid growth of technological advancement in the last few decades due to the contribution of the smart device, which now became an indispensable part of the day-to-day life of humans (Casado et al., 2021). The rapid increase in the demand for mobile devices and the internet increase the usage of the IoT. The gradual increase in IoT devices generates a huge amount of data due to its different types of applications. The IoT data can be gathered from various fields and sources such as smart devices and IoT sensors, and these gathered data are transmitted to the main servers for the evaluation process with the help of different communication strategies like Bluetooth, Wifi, Zigbee, and so forth. The most relevant information is extracted from the IoT data streams through analytical techniques that accept numerous IoT applications (Disabato & Roveri, 2019). There may be some reflection in the data streams due to the influence of data distribution of the collected observation. The domains, which depend on security, monitoring, and control require accurate and fast distribution change detection (Lu et al., 2018; Pratama et al., 2016). The mobile tracking system, which monitors the behaviour of the users and the intrusion detection system which monitors unusual activities (Lu et al., 2020; Zliobaite et al., 2016) are some examples of monitoring and control systems. The variations in the distribution in the data streams are concept drifts in the existing articles. Due to the concept drift presence, the real data is fed forward after carrying the preprocessing steps for the detection and classification system as it degrades the detection and classification accuracy (Xu et al., 2020).

The idea included in the 'concept drift' describes the concealed and unknown interconnection between the input and the output variables (Yuan et al., 2021). Generally, the concept drift is found in various real-time applications that degrade the performance of the system. For example, in the recommender system, the end user preference will be changed concerning financing, trends, and numerous external factors. Furthermore, the climate sensing modules are also altered concerning the seasonal variation in the environment (Patila et al., 2020). These alterations and modifications lead to the degradation of the categorization process. Depending on the speed of the target variables, the concept drift is categorized into gradual and sudden drifts. The sudden or rapid gradual drift is described as the massive variation between the distribution class and the received samples at a particular time frame (Jiwani et al., 2021). The restrained concept drift is a tedious procedure and denotes the variations in various fundamental distributions among new and previous instances. The category of variation is counted and it has to be relevant to view and locate the variations clearly (Priya & Uthra, 2021). Straightforward and ground breaking analyses linked to concept drift have upgraded over the past 10 years. The productive data stream inherits the features of real-time, rapidity, wide-area, and infinite. Based on the time and variation in speed, concept drift is classified into numerous types, such as incremental or gradual, and abrupt. Abrupt (sudden) describes the change between one idea to a new one that happens suddenly. Incremental describes the change between concepts that occurs over time, as the new concept generates. The concept drift issues are more effective for data mining and machine learning (Priya & Uthra, 2021) since more data are stacked together to develop the data streams rather than arranged in static databases. Additionally, deep CNN offers a productive dense network that effectively completes predictions, identifications, and so on. The non-adaptive algorithms fail to cope with varying iterations but the adaptive methods can withstand by varying iterations, which is more efficient.

The learning models based on Active detection methods (Bifet & Gavalda, 2007) are handled by monitoring the changes in the performance detector. The triggering alarm for the concept drift is used to generate an advanced classifier and to restore the conventional classifier, where performance degradation is detected. The advanced ensemble learning models depended on active detection increase in chunk size and enable the presence of noise resulting in fluctuation in the performance. Online ensemble classifier based on learning frameworks upgrades the conventional classifier dynamically and steadily regulate weights regarding the execution. To achieve an effective classification between stability and adaptability historical information is utilized that attenuates the weights of deep learning classifier is invaluable due to the lack of handling the concept drifts (Guo et al., 2021). Additionally, it is expensive and consumes more time while learning. Alternate ways for operating the MI framework without training such as lazy learning, K-nearest neighbour (KNN), case-based reasoning (Homem et al., 2020), and the Bayesian model (Mendez et al., 2006) do not efficiently elaborate the comprehension structure in case of data management hence need the tuning of attributes and data. In case an undetermined value is present in the newly extracted data, they are grouped easily by finding the similarities and variations between the remaining data. Still, they suffer in the classification process where a large amount of data consumes more time (Manning et al., 2008; Priya & Uthra, 2020).

3 | PROPOSED METHODOLOGY FOR HANDLING THE CONCEPT DRIFT

The ultimate goal is to develop a dynamic IoT streaming analytics that handles the issues of online streaming data for concept drift. The proposed concept drift model is comprised of two phases, such as offline learning and incremental learning for drift detection. The proposed optimized

14680394, 2023, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/exsy.13394 by Universidad del Bio Bio, Wiley Online Library on [18/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

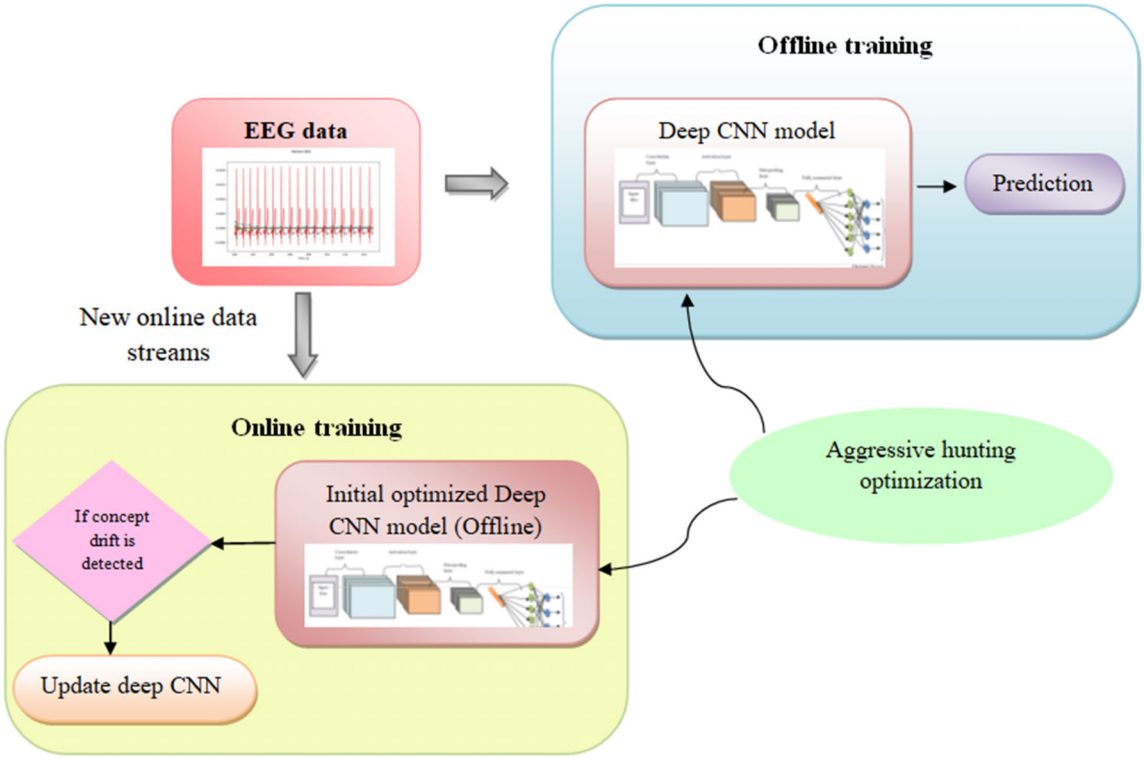


FIGURE 1 Diagrammatic representation of the adaption framework using optimized deep learning for detection.

Deep CNN diagrammatic approach is shown in Figure 1. Offline learning is to attain the initial training model, while incremental learning is utilized to determine the attacks/variants in the online data streams. In the offline learning phase, the real-time IoT data is gathered to create the historical dataset, which is used to train the Deep-CNN model. The hyperparameters of the Deep-CNN model are effectively tuned by the proposed AHO algorithm, a meta-heuristic algorithm that develops the optimized Deep-CNN model is used. In offline training, the already detected attacks are identified using the AHO algorithm and in online training, the new attacks are identified and these attacks get updated in the classifier.

In the incremental learning phase, the continuously generated data streams are processed to determine the concept drifts in online streaming. At the starting state, the trained base classifier is utilized to process the data streams. If the concept drift is determined through the base classifier named proposed AHO algorithm-based deep CNN classifier then, the classifier is retrained for the new sample data collected by the AHO algorithm to befit for the current concept drifts. Thus, the proposed concept drift prediction model adapts to the constantly changing IoT traffic data samples, which provides the most accurate detection.

3.1 | Development of the base classifier stage

In this stage, there are three steps including the feature extraction, feature representation, and base classifier development steps. Let us take a deep insight into the steps.

3.1.1 | Feature extraction and representation step

In this step, the features extracted, which are used to represent the classifier for training purposes, and the features are extracted through the static analysis of the data. The static data is selected because the variations do not occur on the data, which helps in improving the performance of the detection process using the AHO algorithm. In offline training, the relevant features, such as signal power spectrum, amplitude RMS, variance, kurtosis, skewness, min-amplitude, max-amplitude, number of zero crossing, number of local extrema or peak, Petrosian Fractal dimension, Higuchi fractal dimension, total power, Hurst exponent, median frequency, peak frequency, Hijroth parameters, and spectral entropy, are extracted from the input EEG signal and these features are incorporated to form the feature vector, F_v that forms the input to the base classifier, and engages in abnormality detection.

14680394, 2023, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/exsy.13394 by Universidad del Bio Bio, Wiley Online Library on [18/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

3.1.2 | Developing the offline deep CNN classifier

In this research, the base classifier employed is the proposed method, which is developed offline through the integration of the AHO algorithm with the standard deep CNN classifier. The proposed model captures the data variants as the proposed classifier is effective in extracting the behavioural patterns of the malicious samples from the input data. The feature vector is used to train the offline classifier, which is represented as F_v. Hence, the training sample is represented as,

$$\mathrm{ST}_{\mathrm{data}} = \{(\nu^{\tau}, \kappa^{\tau});\ \tau \in (1,2,\ldots,v_{tt})\}, \tag{1}$$

where, τ is the data size and $\mathrm{ST}_{\mathrm{data}}$ is the sample training data, where holds the sample input ν^{τ} and the label κ^{τ}. The trained model is represented as,

$$\mathrm{TM} = \mathrm{Train}[\mathrm{ST}_{\mathrm{data}}]. \tag{2}$$

Following is the detailed architectural explanation of the proposed AHO algorithm-based deep CNN classifier.

I. Optimized deep-CNN model: Deep convolutional neural networks are mostly used for multidimensional signal processing (CNNs). The phrase-deep refers to approaches for teaching these systems to automatically learn their parameters using data types of a particular problem domain of interest and refers generally to networks having from a few to several dozen or more convolution layers. The Deep-CNN is an efficient deep-learning model that automatically detects the most relevant features from the input data and helps in predicting the concept drift in the streaming data. To enhance the practical performance of deep learning, deep CNNs are frequently paired with pooling a limited number of fully linked layers, and other approaches. The Deep-CNN classifier is found to be more effective than some conventional classifiers on the high-dimensional data due to downsampling in the pooling layer. The down-sampling process helps to remove the data with a small gradient during the sample training process, which effectively minimizes memory usage and time. Furthermore, the convolution layer reduces the feature size without losing the most significant information in the data. The architecture of the deep CNN classifier is shown in Figure 2. The parameters of the deep CNN classifier is interpreted in Table 1.

a. Convolution layer: A convolutional layer develops the feature map by applying the filter to the input. The convolution converts all the features in the receptive field to minimize data size. The convolution is the linear operation, which includes the multiplication of the weight with input signals. In the proposed drift prediction model two conventional layers are used one with 64 filters and the other with 32 filters. The final output of the convolutional layer is the feature vector. Let us consider F_v^L as the feature map of the layer L, layer weight W and bias as B, which is mathematically expressed as

$$\mathrm{Oc} = B + WF_v^L. \tag{3}$$

b. Activation layer: Enhances the nonlinearity in the output sample signal. The Rectified Linear Unit (ReLU) layer is the general activation function used in the convolutional layer. The ReLU function is mathematically represented as

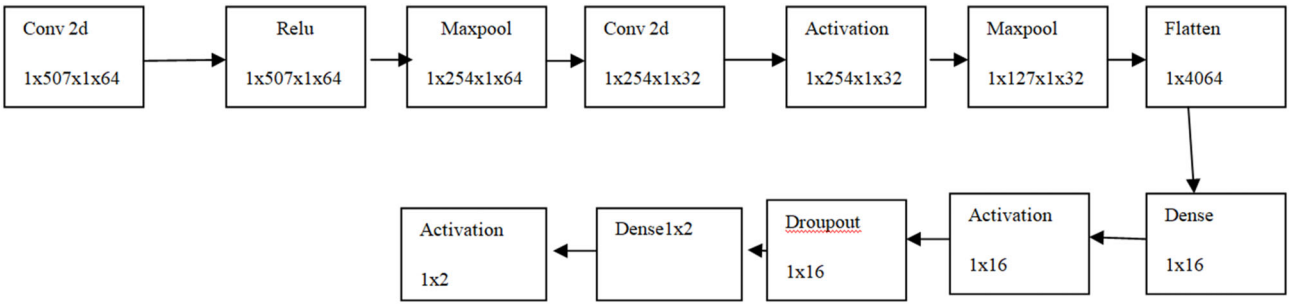


FIGURE 2 Architecture of the proposed deep CNN classifier.

14680394, 2023, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/exsy.13394 by Universidad del Bio Bio, Wiley Online Library on [18/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1 Parameters of deep CNN.

Layer	Output shape	Param #
conv2d	(None, 507, 1, 64)	640
activation	(None, 507, 1, 64)	0
max_pooling2d	(None, 254, 1, 64)	0
conv2d_1	(None, 254, 1, 32)	18,464
activation_1	(None, 254, 1, 32	0
max_pooling2d_1	(None, 127, 1, 32)	0
flatten	(None, 4064)	0
dens	(None, 16)	65,040
activation_2	(None, 16)	0
dropout	(None, 16)	0
dense_1	(None, 2)	34
activation_3	(None, 2)	0

$$\max(0.0, Oc). \tag{4}$$

The above equation represents that if the input value is positive, that value is returned, otherwise, a value of 0.0 is returned.

c. Max-pooling layer: Generally the pooling layer is utilized to minimize the input size of the signal to increase the computational speed and the robustness of the detected features. The pooling layer minimizes computation in the network along with its parameters. The pooling layer operates indirectly on the feature map that minimizes the dimension of the input signals. The max-pooling layer is generally utilized for pooling operation, which helps to select the maximum elements from the feature map size. Hence max-pooling layer output is the feature map that is comprised of the most distinguished features of the previous maps.
d. Fully connected layer: This is utilized to perform the classification based on the features extracted from the previous layer.

II. The proposed Desale's aggressive hunting optimization: The proposed AHO algorithm is the meta-heuristic algorithm, in which a new search agent is developed by integrating the influential hunting characteristics of Sealion (Masadeh et al., 2019) and the predominance rule of the Coyote (Pierezan & Coelho, 2018). Optimizers are programs or techniques that modify the weights and learning rates of the neural network that minimize losses. In deep learning, optimization is applied to tune the hyperparameters in the classifier.

a. Motivation

The sea lion is listed as one of the intellectual animals as it is highly aware of the movement of its prey (fish). The sea lion possesses the capacity to locate its prey and provides an immediate reaction to the prey. They can track out their prey even in shallow water, the surface of the ocean, and near the shore. Moreover, sea lions can move rapidly and constructively in the water. They can chase the prey with high velocity (30 mph) and they can quickly alter their position with the help of their whiskers. On the other hand, the coyote is the social primate that prefers to reside in the troop, the main characteristic of this coyote is that it follows the social hierarchy during the hunting process. Every troop has a leader and the leader will lead the troop according to the information gathered from the other troop members. The troop members are allowed to gather information from different locations. The leader selects the best position, with respect to the best position based on the distance between the search agent and the prey. Hence, in the proposed AHO algorithm the new search agent inherits the unique hunting strategies and also follows the social hierarchy to attain the best prey. These hybrid characteristics of the new search agent avoid the local optima as it provides a tradeoff between exploration and exploitation space and it obtains the global optimal solution within fewer time frames.

b. Mathematical model: The social hierarchy, chasing, attacking, and encompassing prey is elucidated in this section. The proposed AHO algorithm is elucidated as:

14680394, 2023, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/exsy.13394 by Universidad del Bio Bio, Wiley Online Library on [18/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

1. Recognizing and tracking phase: The search agents effectively detect the shape, size, and position of the prey, which helps in the identification of the target variable in the model. If the position of the search agent is identified it provides information to all the members of the group to chase and obtain the prey. This indicates that effective communication is performed in the model to analyse the information and obtaining the prey indicates the recognition of the target variable. There is one controller in every troop, and the group members have to update their position about the prey to the controller. The prey is determined by the search agent and the prey distance. This behaviour is mathematically represented as,

$$\vec{d} = \left| 2\vec{\mathfrak{R}} \cdot \vec{P^t_{\text{tar}}} - \vec{P^t_{\text{Ag}}} \right|. \tag{5}$$

The $\vec{d}$ specifies the distance between the prey and the search agent, the search agent and the target prey position vectors are denoted by $\vec{P^t_{\text{Ag}}}$ and $\vec{P^t_{\text{tar}}}$. Current iteration is denoted as an attribute t and $\vec{\mathfrak{R}}$ denotes the random vector in [0, 1] and the random vector is multiplied by two that enhances the search space, which assists them to obtain the optimal solution. The random vector is assigned for obtaining the desired outcome and adjusted according to the situation. The search agents move closer to the target prey, which is mathematically represented as,

$$\vec{P^{t+1}_{\text{Ag}}} = \vec{P^t_{\text{Ag}}} - \vec{d} \cdot \vec{C}, \tag{6}$$

where, P^{t+1}_{Ag} denotes the search agent during the next iteration $t+1$ and the $\vec{C}$ denotes the movement of the target prey that generally decreased from 2 to 0 for each course of an iteration and it denotes that the search agent is moving toward the prey. This decrease in $\vec{C}$ impels the controller to shift toward the current target and encircle them.

2. Declaration phase: It is already stated that the search agents communicate with each other to share the location, size, and shape of the target. This denotes the characteristics of the target variable present in the model. The controller determines the target and calls other troop members to encompass and attack the target. This declaration behaviour is mathematically modelled as,

$$\vec{\text{Con}_{\text{Speed}}} = \left| \left(\vec{\nu_{\text{water}}} \left(1 + \vec{\nu_{\text{air}}}\right) / \vec{\nu_{\text{air}}} \right) \right|, \tag{7}$$

where, $\vec{\nu_{\text{water}}}$ and $\vec{\nu_{\text{air}}}$ demonstrates the speed of the sound of search agents in water and air respectively. The $\vec{\text{Con}_{\text{Speed}}}$ demonstrates the speed of the controller. The search agents have the capability to communicate both in air and water, which helps the controller to detect the concept drift in varying situations.

3. Exploitation phase: The target prey position is identified by the search agent and encloses the target. The hunting process is directed by the best search agent, who determines the prey and informs all the other members. In the proposed algorithm, the target is considered the best candidate solution. However, new search agents are formed to find out better prey and encompass them. The two phases included in this algorithm to mathematically design the hunting behaviour of the search agents is described below.

a. Diminishing encompassing phase: This characteristic mainly depends on $\vec{C}$ represented in (6). The $\vec{C}$ is reduced linearly from 2 to 0 for the path of search agents during various iterations. The reduction in the value directs the controller of the search agents to proceed toward the prey and encompass them. Thus the current search agent location is determined by the location of the previous best search agent and the current best search agent.
b. Circle upgrading phase: The search agents hunting behaviour is mathematically represented as

$$\vec{P^{t+1}_{\text{Ag}}} = \left| \vec{P^t_{\text{Ag}}} - \vec{C} \right| \cdot \cos(2\pi\mathfrak{R}_3) + \vec{P^t_{\text{Ag}}}, \tag{8}$$

where, the search agent and the targeted prey distance is demonstrated as $\left| \vec{P^t_{\text{Ag}}} - \vec{C} \right|$, $\mathfrak{R}_3$ indicates the random number between $[-1, 1]$ and $\|$ represents the absolute value. The best search agent or controller is determined by the social hierarchical characteristics of the search agents. To find the best position of the search agents for prey, the communicative behaviour of the coyotes is used, where the best position could be identified through effective information shared between them. The best controller position is updated by using,

$$C = C^t_{\text{best}} + \mathfrak{R}_4 . w_C + \mathfrak{R}_5 . w_{\text{troop}}, \tag{9}$$

where, w_C and w_{troop} denotes the weight of the controller and the pack influence. The information in the coyotes is transmitted through one another through packs and each pack consists of certain 7–8 members. The members of the pack leave the group and join another group and hence information is shared with the packs by the newly joined coyotes and this behaviour is represented as the pack influence. The C^t_{best} denotes

14680394, 2023, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/exsy.13394 by Universidad del Bio Bio, Wiley Online Library on [18/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

the previous best solution of the controller. The weight parameters get optimally tuned for deriving the best solution. The attributes $\mathfrak{R}_4$ and $\mathfrak{R}_5$ represents the random number within the span of [0, 1] generated within a uniform distribution. Substituting Equation (9) in (6),

$$P_{\text{Ag}}^{t+1} = \left| P_{\text{Ag}}^{t} - \left(C_{\text{best}}^{t} + \vec{\mathfrak{R}_4}.w_C + \mathfrak{R}_5.w_{\text{troop}}\right) \right|.\cos(2\pi\mathfrak{R}_3) + \vec{P_{\text{Ag}}^{t}}. \tag{10}$$

Equation (8) is the final updated equation, which is formed by the hybridizing hunting characteristic of the coyote and the sea lion. The $\vec{C}$ is updated with the best optimal solution while determining the prey that provided the optimal best position of the search agents.

Termination phase: Until to reach the best global solution with maximum iteration the above procedure is repeated. The proposed model pseudo-code is illustrated in Algorithm 1.

Algorithm 1. Proposed Desale's aggressive hunting algorithm.

S. no.	Pseudo code of the proposed optimization
1	Input: P_{tar}^{t} and P_{Ag}^{t}
2	Output: P_{Ag}^{t+1}
3	Initialize the position of the targeted prey and the search agents.
4	Estimate the distance vector between the prey and the search agent $\vec{d}$.
5	Update the position of search agent $\vec{P_{\text{Ag}}^{t+1}}$
6	Estimate the sound speed of the search agents $\vec{\text{Con}_{\text{Speed}}}$
7	Upgrade the encompassing phase of the search agents
8	Find out the best controller position C
9	The position of the best controller is given as $P_{\text{Ag}}^{t+1} = \left\| P_{\text{Ag}}^{t} - \left(C_{\text{best}}^{t} + \vec{\mathfrak{R}_4}.w_C + \mathfrak{R}_5.w_{\text{troop}}\right) \right\|.\cos(2\pi\mathfrak{R}_3) + \vec{P_{\text{Ag}}^{t}}$.
10	Return global best solution P_{Ag}^{t+1}

3.2 | Incremental learning for concept drift detection

Once the base classifier is trained, the testing phase progresses to detect the concept drift in the system using the online streaming data. Let us denote the test data as, TD, which is given by,

$$\text{TD} = \{(\nu^{\tau}, \kappa^{\tau});\ \tau \in (1,2,...,\text{DD})\}, \tag{11}$$

where, DD corresponds to the total test data, and the predicted class is given by,

$$\hat{\kappa^{\tau}} = \text{TM}[\nu^{\tau}]. \tag{12}$$

Then, the loss function is evaluated between the predicted class and the actual class such that $\forall \text{DD} E^{\tau} = \hat{\kappa^{\tau}} - \kappa^{\tau};\ \kappa \in [0,1]$. During the testing phase, when there is no concept drift in the system, the probability of the output class to the test data is fixed in such a way that the trained model remains relevant for managing the online streaming data. When there is a concept drift, the classifier performance drops imposing the need to learn the variants of the incoming streaming data. Hence, the concept of incremental learning comes into existence and there are several incremental learning approaches. Most of the methods consume larger time and hence, the concept drift is managed online using the OASW approach described below.

3.2.1 | Drift identification using OASW

The OASW algorithm is proposed in this research to identify and adapt the IoT ever-changing data streams for accurate evaluation. The OASW is developed by integrating the ideas of sliding and adaptive windows-based techniques. The two different types of windows are utilized in OASW:

14680394, 2023, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/exsy.13394 by Universidad del Bio Bio, Wiley Online Library on [18/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

(1) A sliding window and the adaptive window. The adaptive window is utilized to store the new samples and to find out the concept drifts in the online streaming data the sliding window is utilized. Let us assume the sliding window size S and the adaptive window maximum size is represented as S'. The drift level and the warning level are denoted by the two thresholds such as A and B for the concept drift detection. The strategies involved in the OASW are as follows. For every incoming data representation d in the streams, the sliding window SW consists of $i-n$ to ith representations. The accuracies of the current window SW_i and last window SW_{i-n} are estimated and calculated. The warning level is attained if there is a drop of ω% from the time interval $i-n$ to i. The adaptive window begins to gather the samples of the received data as the new sample data. If the accuracy of the window continues to drop the accuracy from ω to ξ% a drift is recognized and the window triggers the drift alarm. Now, the classifier is retrained on the new concept from the data sample that is stored in the adaptive window to update the learner.

The adaptive window continues to collect new data samples to obtain stable and robust learners. The sample data collection process is continued until it satisfies any one of the following constraints: (1) If there is an accuracy drop in the new concept to the warning level ω when compared to the starting point of the drift. (2) When the adaptive window reaches its maximum size S'_{max}, which provides the on-time requirement and memory constraints. The learners are again updated on the samples in the adaptive window to become more vigorous learners. For the detection of concept drift the system is now converted to the normal state. On the flip side, if the accuracy of the sliding window stops increase to the normal level or stop droppings, it is found to be a false alarm.

Hence, the four parameters, such as ω, ξ, S and S'_{max} are considered important hyperparameters that have a direct influence on the performance of the OASW system. Hence, the proposed algorithm is used to tune these main hyperparameters to attain the best compatible learning model, as the proposed algorithm is effective for both the discrete and continuous hyperparameters, which belong to OASW. The detected hyperparameters are now utilized for drift adaptation to develop the optimized design for IoT data analytics.

The OASW attains a low run time and minimum space complexity $C(N)$, where the number of occurrences is represented by N the maximum search time taken for the hyperparameter. The OASW model demonstrates the following advantages.

1. All the other conventional method concentrates on either drift adaption or drift detection, while the OASW model performs both functionalities as it utilizes the adaptive window for the adaptation of drifts and the sliding window is utilized for the detection of drifts.
2. An OASW method provides better adaptability and capability than the other conventional methods when the method is applied for independent tasks and datasets, as it tunes the hyperparameter to fit it with specific datasets by utilizing the proposed algorithm.
3. The OASW models detect the drifts in the concept and also upgrade the learning model concerning the degradation performance of the model, this ensures that the learner is upgraded at the time necessary.
4. This proposed model provides the trade-off between the computational complexity and the accuracy by utilizing the adaptive window to gather adequate concept samples while eliminating the older samples.

4 | EXPERIMENTAL WORK AND RESULTS

This result elucidates the results obtained from the proposed concept drift determination and adaptation method is illustrated in this section.

4.1 | Experimental setup

The proposed concept drift determination and the adaptation model are implemented in the python lab with windows 10 operating system, 8 GB RAM, and ROM with more than 100 GB. Database 1 and Database 2 are the two main datasets utilized.

4.2 | Dataset description

a. Database 1: This Database 1 (Shoeb, 2010) is gathered from the children's hospital Boston and this dataset is comprised of EEG recordings from individuals, who are suffering from uncontrollable seizures. The individuals are observed for several days after the retraction of the anti-seizure treatments to categorize their seizures and determine the individual for the surgical process. The total records were arranged into 23 cases that are gathered from 22 individuals. Among the 22 individuals, there are five males from the age group 3 to 22 and 17 females from the age group 1.5 to 19. The file named SUBJECT-INFO consists of the age and gender. The file named RECORDS consists of 664.edf files encompassed in this section. RECORDS-WITH-SEIZURES file consists of 129 files, which contain more than one seizure thus the dataset consists of a total of 198 seizures; The starting and ending of each seizure are interpreted in the seizure annotation file that goes along with the files listed in the file that named as records with seizure. The information about the devices utilized for the recording is briefly described in the file named channel-summary.txt.

14680394, 2023, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/exsy.13394 by Universidad del Bio Bio, Wiley Online Library on [18/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

b. Database 2: This database 2 contains the EEG documents of 14 individuals collected from the Neurophysiology and neurology unit of Siena university. There are five females from the age group 20–58 and nine males from the age group 25–71 from 14 individuals. The individuals are monitored with the help of Video-EEG, whose sampling rate is found to be 512 Hz and the electrodes are arranged in the form of an international 10–20 system. Almost all recording consists of 1 or 2 EKG signals. The diagnosis and categorization of epilepsy were established by the clinical expert with respect to the International league after the attentive evaluation of electrophysiological and clinical data. The database is comprised of 14 folders that contain the EEG recordings in the EDF format.

4.3 | Performance metrics

The key parameters utilized using AHO-based Deep CNN methods are accuracy, sensitivity, specificity, *F*1 measure, and precision. The description of the performance metrics are elaborated in this section.

4.3.1 | Accuracy

The measurement of closeness between the estimated value and the standard value. The system's accuracy is mathematically described as

$$A_{\text{ccu}} = \frac{Tr_{\text{pos}} + Tr_{\text{neg}}}{Tr_{\text{pos}} + Tr_{\text{neg}} + \Gamma a_{\text{pos}} + \Gamma a_{\text{neg}}}. \tag{13}$$

4.3.2 | Sensitivity

This is the sum of all true positive values correctly identified by the system. The system's sensitivity is mathematically described as

$$\text{Se}_n = \frac{Tr_{\text{pos}}}{Tr_{\text{pos}} + \Gamma a_{\text{neg}}}. \tag{14}$$

4.3.3 | Specificity

Specificity is the sum of true negative values that the system has correctly identified. The mathematical representation of the system's specificity is

$$\text{Sp}_e = \frac{Tr_{\text{pos}}}{Tr_{\text{pos}} + \Gamma a_{\text{neg}}}. \tag{15}$$

4.3.4 | Precision

The term 'precision' illustrates how closely the two measurements are placed to each other.

$$\text{pr}_e = \frac{Tr_{\text{pos}}}{Tr_{\text{pos}} + \Gamma a_{\text{pos}}}. \tag{16}$$

4.3.5 | *F*1-measure

The *F*1-measure is estimated using the following equation.

$$F1\text{-measure} = \frac{2Tr_{\text{pos}}}{2Tr_{\text{pos}} + \Gamma a_{\text{pos}} + \Gamma a_{\text{neg}}}. \tag{17}$$

14680394, 2023, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/exsy.13394 by Universidad del Bio Bio, Wiley Online Library on [18/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

4.4 | Experimental evaluation

AHO-based Deep CNN method concentrates on the training of the deep CNN classifier using an optimization algorithm. Hence, the datasets are used as binary classification datasets. The frequential and the hold-out validation models are utilized for the evaluation of the models. In the hold-out validation phase, 10% of the sample data are utilized as the training set for initial training. The last 90% of the data are utilized as the test set for online training. The prequential validation phase, which is also termed test and train validation is utilized for the online training, in which the input instance is first utilized for the test model and then is utilized for the updating model.

5 | COMPARATIVE ANALYSIS

5.1 | Comparative methods

Comparative methods (CM) consist of eight models, such as Multi-Layer Perceptron (CM-1) (Pratama et al., 2019), Support Vector Machine (CM-2) (Ratnayake & Amer, 2019), Random Forest (CM-3) (Su & Liu, 2018), AdaBoost (CM-4) (De Carvalho Santos & de Barros, 2020), K-nearest neighbour (CM-5) (Liu et al., 2018), Deep-CNN (CM-6) (Kollias et al., 2017), SLO-based Deep CNN (CM-7) (Kollias et al., 2017; Masadeh et al., 2019), COA-based Deep CNN (CM-8) (Kollias et al., 2017; Pierezan & Coelho, 2018) are utilized in this research.

5.2 | Comparative analysis

After the utilization of AHO, the best hyper-parameters were allocated to the proposed concept drift models to develop the best concept drift, detection models. Figures 3 and 4 illustrate the accuracy comparison of the AHO-based Deep CNN model with the conventional models. From Figure 3, it demonstrated the AHO-based Deep CNN model surpasses all the competent methods in terms of accuracy. While using Database 1, as demonstrated in Figure 3, the proposed model attains a maximum accuracy of 99.76% compared to all the conventional models and a concept drift is determined at the point of 2999.00. On considering the method in the absence of drift adaptation, the offline Deep CNN model

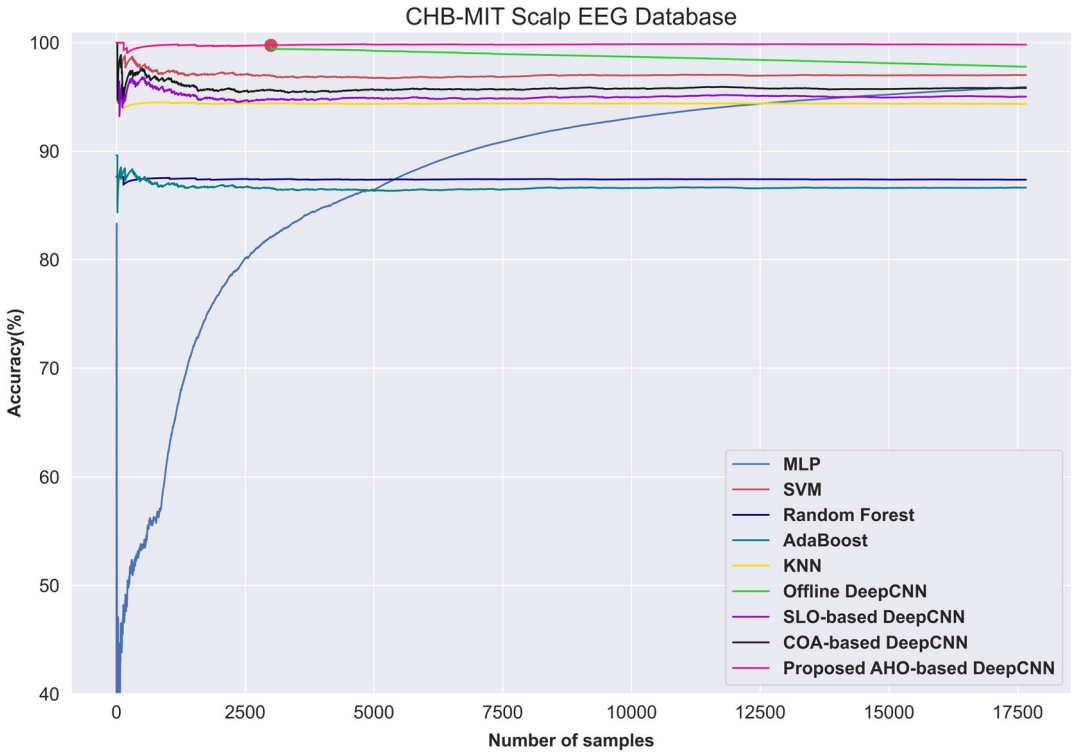


FIGURE 3 Accuracy evaluation of the drift adaptation methods using the CHB-MIT scalp EEG database.

14680394, 2023, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/exsy.13394 by Universidad del Bio Bio, Wiley Online Library on [18/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

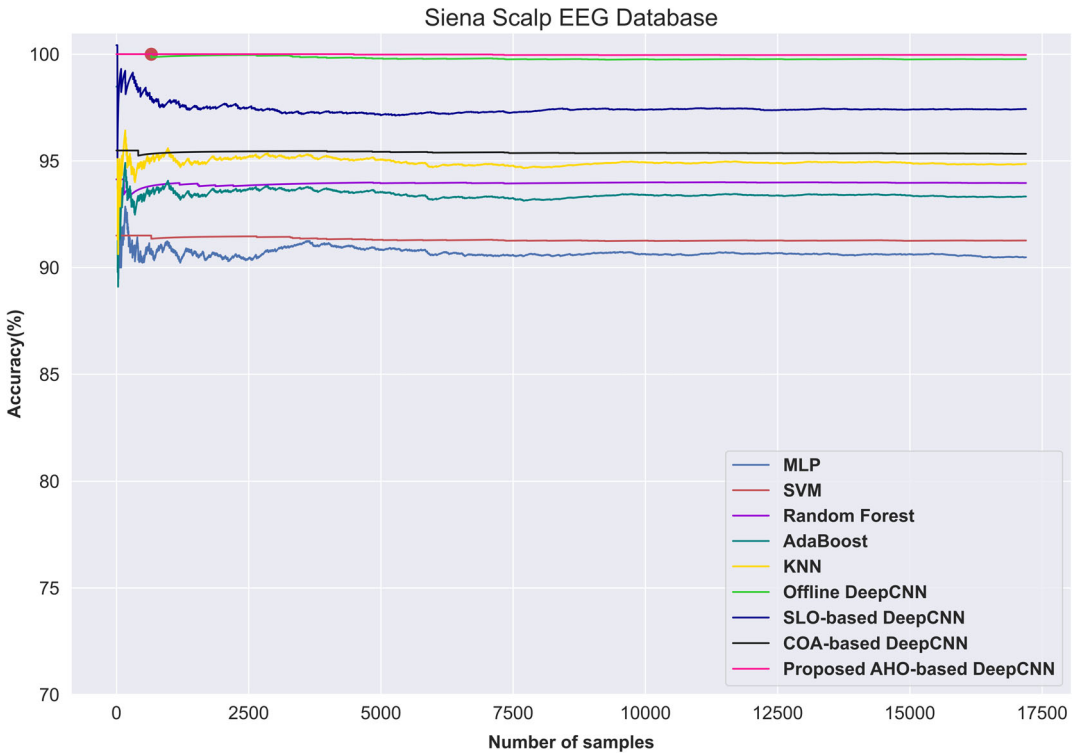


FIGURE 4 Accuracy evaluation of the drift adaptation methods using the Siena scalp EEG database.

provides lower accuracy of 99.75%. The accuracies of the further seven existing methods, which show an accuracy variation between (82.067% and 95.27%) are also comparatively lesser than AHO-Deep CNN method.

For Database 2, there experience a high drift at the initial phase of the test set. As demonstrated in Figure 4, by adjusting to the unexpected drift determined at the point of 983, AHO-Deep CNN model gains the highest accuracy of 99.96%, while the accuracy of the offline Deep-CNN model significantly drops to 99.75%, in the absence of drift adaptation. This enumerates the efficiency of the drift adaptation framework. Eight comparative methods such as MLP, SVM, Random Forest, AdaBoost, K-nearest neighbour, Deep-CNN, and SLO-based Deep CNN, inherit lower accuracy, while compared to the proposed method (90.68%–97.62%).

5.3 | Comparative analysis using CHB-MIT scalp EEG database

This is made using dataset 1 and the metrics accuracy, sensitivity, specificity, *F*1 measure, and precision is measured and is shown in Figure 5. For deeper analysis, the improvement percentage obtained by the AHO-deep CNN classifier is measured. First the measured accuracy values are shown in Figure 5a, where the proposed method obtained an improvement of 13.122% over CM-1, 11.534% over CM-2, 14.326% over CM-3, 14.246% over CM-4, 3.593% over CM-5, 4.808% over CM-6, 4.332% over M-7, 3.162% over M-8 during the training percentage 90. Second, the measured sensitivity values are shown in Figure 5b, where the proposed method obtained an improvement of 11.017% over CM-1, 11.818% over CM-2, 11.470% over CM-3, 11.408% over CM-4, 3.442% over CM-5, 5.796% over CM-6, 2.408% over CM-7, 3.987% over CM-8 during the training percentage 90. Third, the measured specificity values are shown in Figure 5c, where the proposed method obtained an improvement of 12.58% over CM-1, 6.80% over CM-2, 14.61% over CM-3, 11.21% over CM-4, 3.50% over CM-5, 3.64% over CM-6, 1.24% over CM-7, 1.68% over CM-8 during the training percentage 90. Likewise, the measured *F*1 measure is shown in Figure 5d, where the proposed method obtained an improvement of 11.2% over CM-1, 14.6% over CM-2, 17.5% over CM-3, 20.5% over CM-4, 4.9% over CM-5, 2.0% over CM-6, 2.1% over CM-7, 2.7% over CM-8 during the training percentage 90. Finally, the precision value is measured shown in Figure 5e, where the AHO-based deep CNN classifier obtained an improvement of 9.49% over CM-1, 9.87% over CM-2, 11.02% over CM-3, 16.48% over CM-4, 6.64% over CM-5, 6.74% over CM-6, 4.75% over CM-7, 1.19% over CM-8 during the training percentage 90. The improvement percentage obtained by the proposed model over the state of art method proves the significance of the model and the values obtained by the proposed model are interpreted in the table.

14680394, 2023, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/exsy.13394 by Universidad del Bio Bio, Wiley Online Library on [18/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

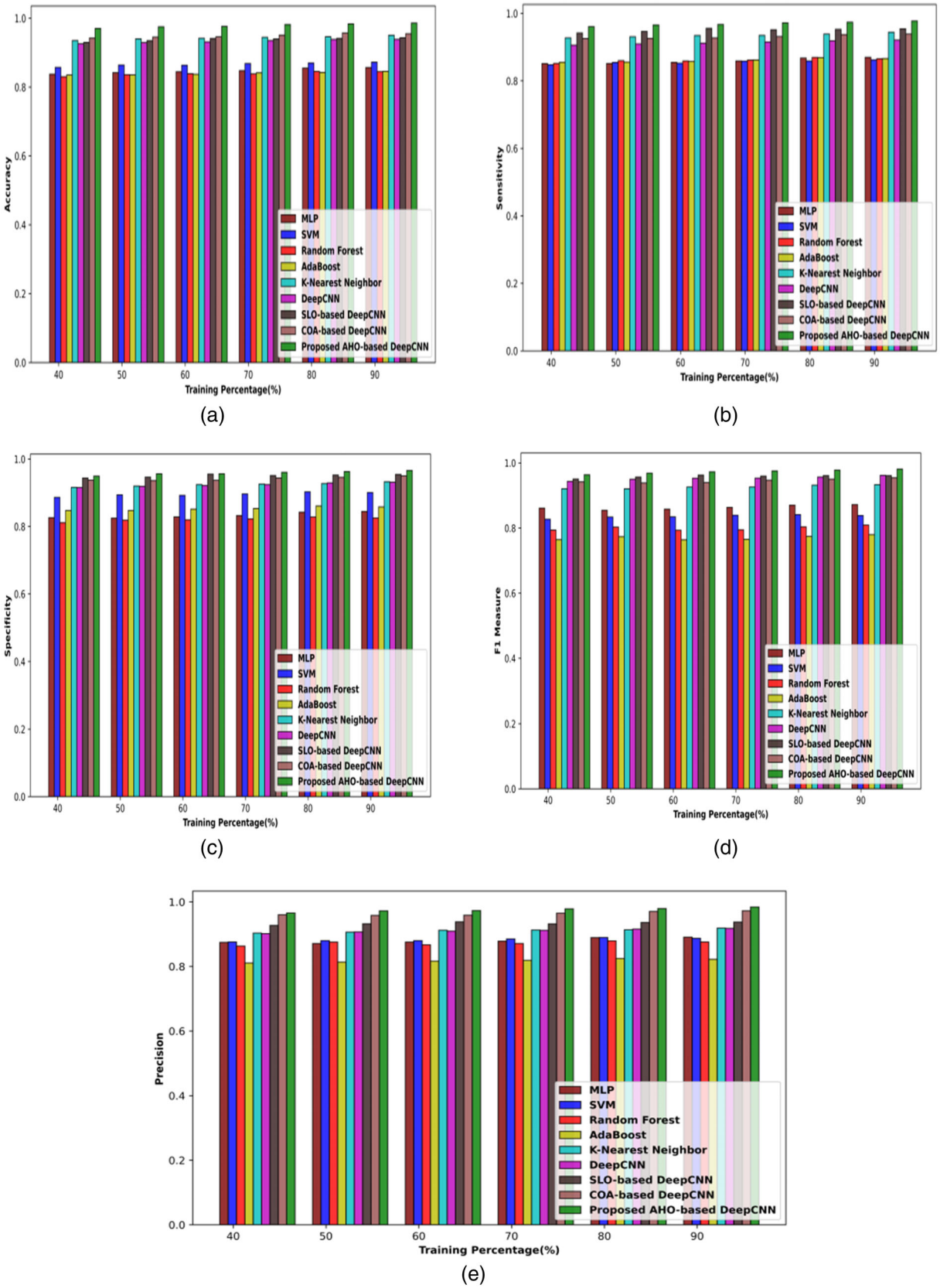


FIGURE 5 Comparative analysis using CHB-MIT scalp EEG database, (a) accuracy, (b) sensitivity, (c) specificity, (d) *F*1-measure, and (e) precision.

14680394, 2023, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/exsy.13394 by Universidad del Bio Bio, Wiley Online Library on [18/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

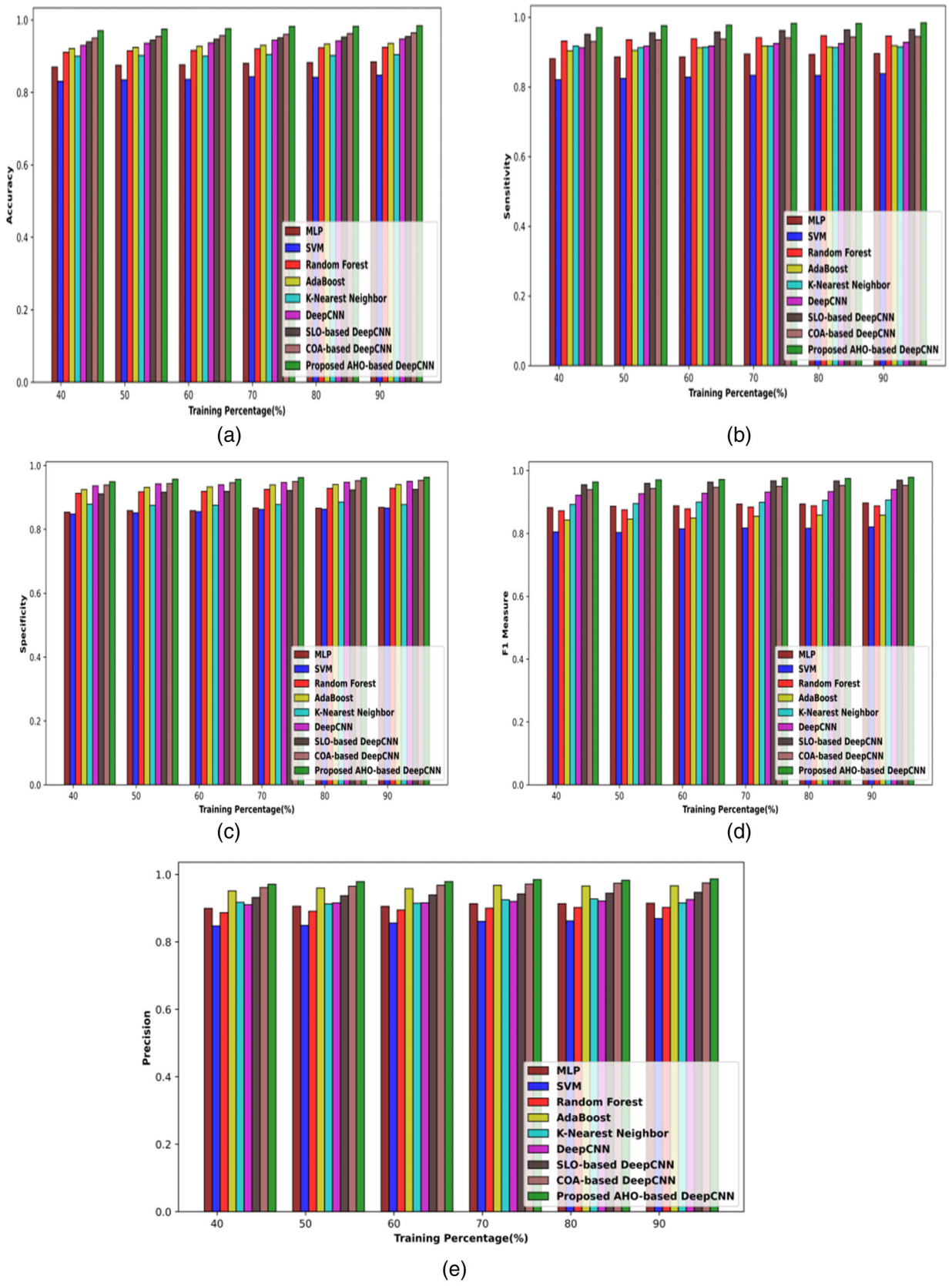


FIGURE 6 Comparative analysis using Siena scalp EEG database, (a) accuracy, (b) sensitivity, (c) specificity, (d) *F*1-measurement, and (e) precision.

14680394, 2023, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/exsy.13394 by Universidad del Bio Bio, Wiley Online Library on [18/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

5.4 | Comparative analysis using Siena EEG database

This is made using dataset 2 and the metrics accuracy, sensitivity, specificity, *F*1 measure, and precision are measured and is shown in Figure 5. For deeper analysis, the improvement percentage obtained by the proposed method is measured.

First the measured accuracy values are shown in Figure 6a, where the proposed AHO-based deep CNN classifier obtained an improvement of 9.76% over CM-1, 13.86% over CM-2, 6.03% over CM-3, 4.93% over CM-4, 8.11% over CM-5, 3.7 6% over CM-6, 3.03% over CM-7, 2.01% over CM-8 during the training percentage 90. Second, the measured sensitivity values are shown in Figure 6b, where the proposed method obtained an improvement of 8.99% over CM-1, 14.86% over CM-2, 3.86% over CM-3, 6.66% over CM-4, 7.01% over CM-5, 5.70% over CM-6, 1.94% over CM-7, 4.01% over CM-8 during the training percentage 90. Third, the measured specificity values shown in Figure 6c, where the proposed method obtained an improvement of 9.81% over CM-1, 10.00% over CM-2, 3.55% over CM-3, 2.39% over CM-4, 8.87% over CM-5, 1.35% over CM-6, 3.95% over CM-7, 0.99% over CM-8 during the training percentage 90. Likewise, the *F*1 measure is measured shown in Figure 6d, where the AHO-based deep CNN classifier obtained an improvement of 8.35% over CM-1, 16.14% over CM-2, 9.30% over CM-3, 12.29% over CM-4, 7.40% over CM-5, 3.98% over CM-6, 0.94% over CM-7, 2.61% over CM-8 during the training percentage 90. Finally, the measured precision values are shown in Figure 6e, where the proposed method obtained an improvement of 7.61% over CM-1, 12.20% over CM-2, 8.83% over CM-3, 2.36% over CM-4, 7.52% over CM-5, 6.49% over CM-6, 4.35% over CM-7, 1.51% over CM-8 during the training percentage 90.

5.5 | Convergence analysis

This is performed by measuring the loss that occurred during the various training percentage and is shown in Figure 7a,b. For simplicity, the loss occurred by the methods CM-6, M-7, and CM-8 the proposed method is measured, which obtained the values of 6.11%, 5.65%, 4.49%, and 1.37% for database 1. Similarly, the loss occurred by the methods for database 2 is measured and the loss occurred is 5.28%, 4.56%, 3.56%, and 1.58%. The observation shows that the proposed classifier obtained less loss and is considered more efficient.

5.6 | Comparative discussion

The comparative discussion of the conventional and the proposed method is elaborated in this section. The improved characteristic is obtained by including the optimization which helps the proposed method to gain the highest accuracy compared to some previous existing methods. The significant advantage of Desale's aggressive hunt optimization is its position update behaviour using the enhanced communication characteristics, through the effective information shared between them. Due to the addition of this communication characteristic, the proposed method becomes more efficient and provided better results. The comparative evaluations are accomplished by evaluating the parameters, such as accuracy, sensitivity, specificity, *F*1-measure, and precision using the datasets, such as the seizure recognition dataset, CHB-MIT Scalp EEG Database, and Siena

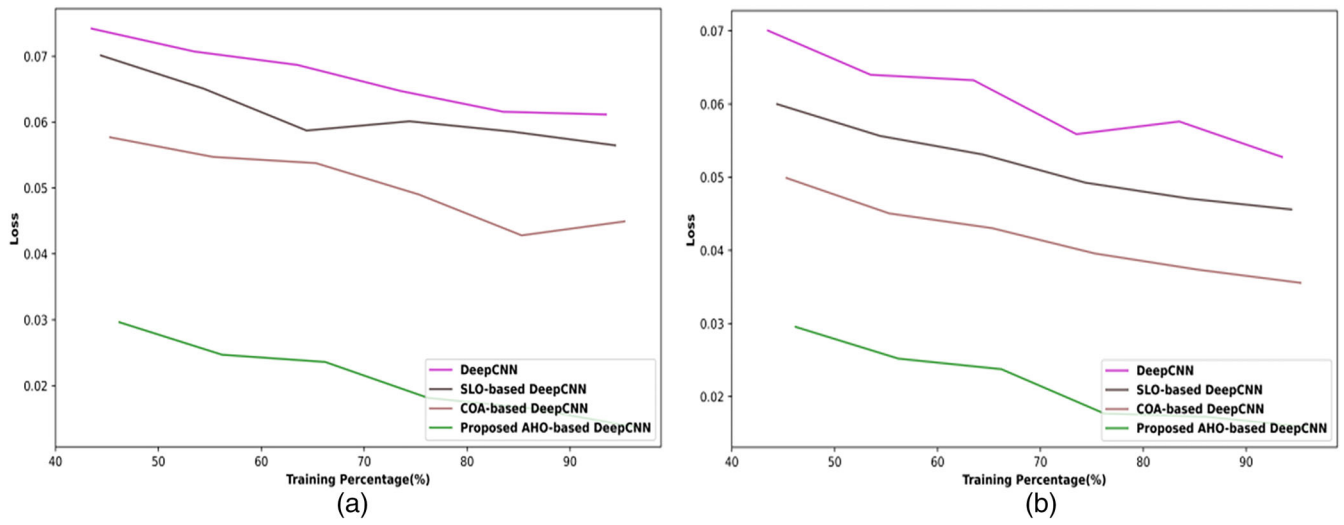


FIGURE 7 Convergence analysis based on (a) database 1 and (b) database 2.

14680394, 2023, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/exsy.13394 by Universidad del Bio Bio, Wiley Online Library on [18/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2 Comparison table based on the performance measures.

Methods	CHB-MIT scalp EEG database			Siena scalp EEG database		
	Accuracy (%)	Sensitivity (%)	Specificity (%)	Accuracy (%)	Sensitivity (%)	Specificity (%)
MLP	85.69	87.00	84.50	88.4357	86.9075	89.7205
SVM	87.25	86.21	90.08	84.7774	86.7166	82.0952
RF	84.50	86.55	82.53	92.4907	92.9391	88.7931
AdaBoost	84.58	86.61	85.82	93.5680	94.0539	85.8673
KNN	95.08	94.40	93.27	90.4400	87.8119	90.6530
Deep-CNN	93.89	92.10	93.13	94.7244	95.0544	94.0012
SLO-Deep LSTM	94.35	95.41	95.45	95.4417	92.5480	96.9760
COA-Deep CNN	95.51	93.87	95.02	96.4450	95.3972	95.3478
Proposed Desale's AHO-Deep CNN	98.63	97.77	96.65	98.4214	96.3559	97.8990

Scalp EEG Database. Table 2 demonstrates the comparative discussion of the methods for 90% of training. The accuracy of the proposed model is observed as 98.63% and 98.4214%, respectively for the two datasets with 90% of training, which shows that the proposed method outperforms all the other existing drift adaption methods.

6 | CONCLUSION AND FUTURE WORKS

A method for handling the concept drift issues in online data streaming is proposed in this research, which involves offline learning and incremental learning. This research develops the dynamic streaming data analytic framework based on the optimized Deep CNN classifier and OASW approach that effectively addresses both memory and time constraints. The concept drift in the online data streams is effectively predicted using the proposed approach and the proposed optimized Deep CNN classifier as a base classifier establishes the trained model for handling the offline learning process. The effectiveness of the research relies on the proposed nature-inspired algorithm known as the Desale's aggressive hunt optimization algorithm, as it effectively trains the classifier to boost its performance. Moreover, an optimized adaptive and sliding window (OASW) is utilized in this research to adapt the pattern changes in the data streams during incremental learning. The experimental analysis with the comparative methods based on the performance metrics is found to be effective with accuracy, sensitivity, specificity, *F*1 score, and precision of 98.63%, 97.77%, 96.65%, 98.1487%, and 98.4469%, respectively. In future work, the ensemble classifier will be employed to train the more complex data. The optimization techniques based on different classifiers could help further enhancement of the results.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

ORCID

Ketan Sanjay Desale https://orcid.org/0000-0002-2859-5515
Swati V. Shinde https://orcid.org/0000-0002-8271-3025

REFERENCES

Abbasi, A., Firouzi, B., & Sendur, P. (2021). On the application of Harris hawks optimization (HHO) algorithm to the design of microchannel heat sinks. *Engineering with Computers*, *37*, 1409–1428.

Bifet, A., & Gavalda, R. (2007). Learning from time-changing data with adaptive windowing. In *Proceedings of the 2007 SIAM international conference on data mining*, Minneapolis, United States (pp. 443–448).

Casado, F. E., Lema, D., Criado, M. F., Iglesias, R., Regueiro, C. V., & Barro, S. (2021). Concept drift detection and adaptation for federated and continual learning. *arXiv preprint arXiv:2105.13309*.

Darem, A. A., Ghaleb, F. A., Al-Hashmi, A. A., Abawajy, J. H., Alanazi, S. M., & Al-Rezami, A. Y. (2021). An adaptive behavioral-based incremental batch learning malware variants detection model using concept drift detection and sequential deep learning. *IEEE Access*, *9*, 97180–97196.

De Carvalho Santos, S. G. T., & de Barros, R. S. M. (2020). Online AdaBoost-based methods for multiclass problems. *Artificial Intelligence Review*, *53*(2), 1293–1322.

Demsar, J., & Bosnic, Z. (2018). Detecting concept drift in data streams using model explanation. *Expert Systems with Applications*, *92*, 546–559.

14680394, 2023, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/exsy.13394 by Universidad del Bio Bio, Wiley Online Library on [18/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Disabato, S., & Roveri, M. (2019). Learning convolutional neural networks in presence of concept drift. In *2019 international joint conference on neural networks (IJCNN)* (pp. 1–8).

Guo, H., Zhang, S., & Wang, W. (2021). Selective ensemble-based online adaptive deep neural networks for streaming data with concept drift. *Neural Networks*, *142*, 437–456.

Haug, J., & Kasneci, G. (2021). Learning parameter distributions to detect concept drift in data streams. In *2020 25th international conference on pattern recognition (ICPR)* (pp. 9452–9459). IEEE.

Homem, T. P. D., Santos, P. E., Costa, A. H. R., Bianchi, R. A. C., & Mantaras, R. L. (2020). Qualitative case-based reasoning and learning. *Artificial Intelligence*, *283*, 103258.

Jiwani, N., Gupta, K., & Whig, P. (2021). Novel HealthCare framework for cardiac arrest with the application of AI using ANN. In *In 2021 5th international conference on information systems and computer networks (ISCON)* (pp. 1–5). IEEE.

Kollias, D., Yu, M., Tagaris, A., Leontidis, G., Stafylopatis, A., & Kollias, S. (2017). Adaptation and contextualization of deep neural network models. In *Proceedings of 2017 IEEE symposium series on computational intelligence (SSCI)* (pp. 1–8).

Korycki, L., & Krawczyk, B. (2020). Adversarial concept drift detection under poisoning attacks for robust data stream mining. *arXiv preprint arXiv: 2009.09497*.

Liu, A., Lu, J., Liu, F., & Zhang, G. (2018). Accumulating regional density dissimilarity for concept drift detection in data streams. *Pattern Recognition*, *76*, 256–272.

Liu, A., Lu, J., & Zhang, G. (2020). Concept drift detection: Dealing with missing values via fuzzy distance estimations. *IEEE Transactions on Fuzzy Systems*, *29*, 3219–3233.

Lu, J., Liu, A., Dong, F., Gu, F., Gama, J., & Zhang, G. (2018). Learning under concept drift: A review. *IEEE Transactions on Knowledge and Data Engineering*, *31*, 2346–2363.

Lu, J., Liu, A., Song, Y., & Zhang, G. (2020). Data-driven decision support under concept drift in streamed big data. *Complex & Intelligent Systems*, *6*(1), 157–163.

Manning, C. D., Raghavan, P., & Schutze, H. (2008). *Introduction to information retrieval*. Cambridge University Press.

Masadeh, R., Mahafzah, B. A., & Sharieh, A. (2019). Sea lion optimization algorithm. *International Journal of Advanced Computer Science and Applications*, *10*(5), 388–395.

Mendez, J., Fdez-Riverola, F., Díaz, F., Iglesias, E., & Corchado, J. (2006). Tracking concept drift at feature selection stage in SpamHunting: An anti-spam instance-based reasoning system. In *Proceedings of the 8th European conference on case-based reasoning*.

Nagar, P., Khemka, M., & Arora, C. (2020). Concept drift detection for multivariate data streams and temporal segmentation of daylong egocentric videos. In *Proceedings of the 28th ACM international conference on multimedia* (pp. 1065–1074).

Patila, S., Chintamania, S., Dennisb, B. H., & Kumarc, R. (2020). *Predicting internal temperature of heat generating cylinder from surrounding coolant temperature measurements*.

Pierezan, J., & Coelho, L. S. (2018). Coyote optimization algorithm: A new metaheuristic for global optimization problems. In *2018 IEEE congress on evolutionary computation (CEC)* (pp. 1–8).

Pratama, M., Lu, J., Lughofer, E., Zhang, G., & Er, M. J. (2016). An incremental learning of concept drifts using evolving type-2 recurrent fuzzy neural networks. *IEEE Transactions on Fuzzy Systems*, *25*(5), 1175–1192.

Pratama, M., Za'in, C., Ashfahani, A., Ong, Y. S., & Ding, W. (2019). Automatic construction of multi-layer perceptron network from streaming examples. In *Proceedings of the 28th ACM international conference on information and knowledge management* (pp. 1171–1180).

Priya, S., & Uthra, R. A. (2020). An effective concept drift detection technique with kernel extreme learning machine for email spam filtering. In *2020 3rd international conference on intelligent sustainable systems (ICISS)* (pp. 774–779).

Priya, S., & Uthra, R. A. (2021). Deep learning framework for handling concept drift and class imbalanced complex decision-making on streaming data. *Complex & Intelligent Systems*, 1–17.

Ratnayake, K., & Amer, M. A. (2019). Drift detection using SVM in structured object tracking. In *Proceedings of international conference on image analysis and recognition* (pp. 67–76).

Shoeb, A. H., & Guttag, J. V. (2010). Application of machine learning to epileptic seizure detection. In *Proceedings of the 27th international conference on machine learning (ICML-10)* (pp. 975–982).

Su, H. Y., & Liu, T. Y. (2018). Enhanced-online-random-forest model for static voltage stability assessment using wide area measurements. *IEEE Transactions on Power Systems*, *33*(6), 6696–6704.

Xu, R., Cheng, Y., Liu, Z., Xie, Y., & Yang, Y. (2020). Improved long short-term memory based anomaly detection with concept drift adaptive method for supporting IoT services. *Future Generation Computer Systems*, *112*, 228–242.

Yu, H., Sun, C., Yang, X., Zheng, S., & Zou, H. (2019). Fuzzy support vector machine with relative density information for classifying imbalanced data. *IEEE Transactions on Fuzzy Systems*, *27*(12), 2353–2367.

Yuan, Y., Wang, Z., & Wang, W. (2021). Unsupervised concept drift detection based on multi-scale slide windows. *AdHoc Networks*, *111*, 102325.

Zliobaite, I., Pechenizkiy, M., & Gama, J. (2016). *An overview of concept drift applications* (Vol. 4, pp. 91–114). Springer.

AUTHOR BIOGRAPHIES

Ketan Sanjay Desale is currently a PhD student at Department of Computer Engineering, PCET's Pimpri Chinchwad College of Engineering (PCCOE), Pune, India. He completed his ME Degree in Computer Networks at D. Y. Patil School of Engineering Y technology, Pune University, Pune between years 2013 and 2015. Between the year 2007 and 2011 he completed his BE Degree in Information Technology from PCET's Pimpri Chinchwad College of Engineering (PCCOE), Pune, India.

14680394, 2023, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/exsy.13394 by Universidad del Bio Bio, Wiley Online Library on [18/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Swati Shinde is currently working as Professor in Department of Computer Engineering, PCET's Pimpri Chinchwad College of Engineering (PCCOE), Pune, India. She completed her PhD in Computer Science & Engineering. She is also working as Principal Investigator - DST Funded Research Project, NVIDIA Ambassador and Certified Instructor for Deep Learning. Her research interest is in Artificial Neural Networks, Intelligent Systems, Deep Learning and Fuzzy Systems.

How to cite this article: Desale, K. S., & Shinde, S. V. (2023). Concept drift detection and adaption framework using optimized deep learning and adaptive sliding window approach. *Expert Systems*, *40*(9), e13394. https://doi.org/10.1111/exsy.13394

14680394, 2023, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/exsy.13394 by Universidad del Bio Bio, Wiley Online Library on [18/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License